

Topic Covered:

- Probability rules

1. If one card is drawn from a deck, find the probability of getting these results.

- A queen.
- A club.
- A queen of clubs.
- A 3 or an 8.
- A 6 or a spade.



Cards (52)			
Spade	Club	Diamond	Heart
1 King	1 King	1 King	1 King
1 Queen	1 Queen	1 Queen	1 Queen
1 Jack	1 Jack	1 Jack	1 Jack
1 Ace	1 Ace	1 Ace	1 Ace
2-10 Cards	2-10 Cards	2-10 Cards	2-10 Cards
Total = 13	Total = 13	Total = 13	Total = 13

Answer:

- $P(\text{queen}) = 4/52 = 1/13$
- $P(\text{club}) = 13/52 = 1/4$
- $P(\text{queen of clubs}) = 1/52$
- $P(3 \text{ or } 8) = 8/52 = 2/13$
- $P(6 \text{ or spade}) = 16/52 = 4/13$

2. Human blood is grouped into four types. The percentages of Americans with each type are listed below. Choose one American at random. Find the probability that this person

- Has type O blood.
- Has type A or B.
- Does not have type O or A.

O: 43%

A: 40%

B: 12%

AB: 5%

Answer:

- $P(O) = 43/100 = 0.43$
- Since blood types are mutually exclusive: $P(A \text{ or } B) = P(A) + P(B) = 0.40 + 0.12 = 0.52$
- $P(\text{not O or A}) = P(B \text{ or } AB) = P(B) + P(AB) = 0.12 + 0.05 = 0.17$
 Alternatively:
 $P(\text{not O or A}) = 1 - P(O \text{ or } A) = 1 - (0.43 + 0.40) = 1 - 0.83 = 0.17$

3. A sales representative who visits customers at home finds she sells 0, 1, 2, 3, or 4 items according to the following frequency distribution. Find the probability that she sells the following.

- Exactly 1 item.
- More than 2 items.
- At least 1 item.
- At most 3 items.

Items sold	0	1	2	3	4
Frequency	8	10	3	2	1

Answer:

- $P(1) = 10/24 = 5/12$
- $P(>2) = P(3 \text{ or } 4) = (2 + 1)/24 = 3/24 = 1/8$
- $P(\text{at least } 1) = 16/24 = 2/3$
- $P(\text{at most } 3) = 23/24$



4. If one card is drawn from an ordinary deck of cards, find the probability of getting the following.
- A king or a queen or a jack.
 - A club or a heart or a spade.
 - A king or a queen or a diamond.
 - An ace or a diamond or a heart.
 - A 9 or a 10 or a spade or a club.

Answer:

- $P(\text{king or queen or jack}) = 12/52 = 3/13$
- $P(\text{club or heart or spade}) = 39/52 = 3/4$
- $P(\text{king or queen or diamond}) = 19/52$
- $P(\text{ace or diamond or heart}) = 28/52 = 7/13$
- $P(9 \text{ or } 10 \text{ or spade or club}) = 30/52 = 15/26$

5. It was reported that 19.8% of computer games sold in 2005 were classified as “family and children’s.” Choose five purchased computer games at random. Find the probability that
- None of the 5 was family and children’s.
 - At least 1 of the 5 was family and children’s.

Answer:

- $P(\text{not a family and children’s game}) = 1 - 0.198 = 0.802$
 $P(\text{none of five are family and children’s game}) = (0.802)^5 = 0.332$
- $P(\text{at least one is family and children’s game}) = 1 - P(\text{none}) = 1 - 0.332 = 0.668$

Alternatively:

using the binomial probability formula: $P(X = k) = C(n, k) p^k (1 - p)^{n-k}$
where $C(n, k)$ is the binomial coefficient, $p = 0.198$, and $n = 5$.

- $P(\text{not a family and children’s game}) = P(X = 0) = C(5, 0) (0.198)^0 (1 - 0.198)^{5-0} = 0.332$
- $P(\text{at least one is family and children’s game}) = 1 - P(\text{none}) = 1 - 0.332 = 0.668$

